

DS380 LLM Culture: ISSS Scambot

General Motivation and Background

Thousands of international students and scholars at Vanderbilt University are targeted by scams that result to the loss of thousands of dollars each year. Recognizing the need for preventative measures, the International Student & Scholar Services (ISSS) office approached me to develop a scam detection chatbot designed to assist students, scholars, and faculty in identifying potential scams and avoiding victimization.

International students are especially vulnerable to scams as factors like language barriers and limited familiarity with U.S. policies make them prime targets. Scammers exploit these vulnerabilities by impersonating official organizations and use tactics like imitating legitimate email addresses, forging official-looking documents, and threatening deportation, fines, or arrest to pressure students into making immediate payments or revealing sensitive personal information.

As an international student myself, I understand the fear and confusion these scams can create. The U.S. policy environment is hard to navigate and guidance is often inaccessible. This project is my way of offering a practical support system to help others navigate safely and confidently during their time in the U.S.

System Design

The ScamBot is a retrieval augmented generation (RAG) chatbot system with conversational memory. It is built on OpenAI's GPT 4.0 model and uses a FAISS vector database to retrieve relevant documents from a curated knowledge base. User interaction data is also logged through the Google Sheets API for anonymous session tracking.

Scam Safety Chatbot

The ISSS Scam Bot is designed to assist you in determining if a situation you experienced—such as a call, email, or offer—is likely a scam. You can input a summary of the interaction, and the bot will evaluate it based on key scam indicators and give an informed opinion on whether or not it appears to be a scam.

Hello! I'm here to help you determine if a call, email, or other message you've received might be a scam. Just share a brief summary of the situation, and I'll analyze it for common scam indicators like suspicious requests, too-good-to-be-true offers, or unusual urgency. Remember, it's always better to be cautious! Let's get started—describe what happened, and I'll guide you through it

i recieved an email from what seems to be immigration and they had said i need to pay a 200 giftcard to maintain my student status in the US? my sevis?

This situation has multiple indicators of a scam:

1. **Gift Card Payment Request:** Legitimate agencies, including immigration services, do not request payments via gift cards.
2. **Threatening Consequences:** The email threatens your student status, creating a sense of urgency and pressure to comply quickly.
3. **Impersonation:** Scams often involve impersonating government agencies or officials. Government agencies, like USCIS or SEVIS, would not demand payments this way.

Please enter your question



Figure 1: User interface of the chatbot on a streamlit platform (will eventually be integrated on the ISSS platform).

When a user submits a query, the bot retrieves contextually relevant text snippets from official ISSS documents and generates a response grounded in those materials. The chatbot maintains session memory to provide context aware replies across a conversation. Each interaction is tagged with a unique session ID and timestamp to help the ISSS office track and analyze patterns of use while preserving user privacy.

ISSS_Responses

File Edit View Insert Format Data Tools Extensions Help

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123

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Figure 2: Response tracking on google sheets

A key feature not demonstrated in the AI Showcase is the internal data ingestion tool. Accessible through a separate internal facing user interface, this tool allows ISSS staff to upload and manage documents that populate the vector store. By giving staff direct control over the source material, ScamBot can generate responses with the most current scam tactics and policy updates.

Designed for integration into the Vanderbilt ISSS website, ScamBot provides international students with a trusted, easily accessible resource for scam awareness and prevention.

Internal: Upload or Paste Materials for the ISSS ScamBot

Welcome! Please upload .txt or .html files OR fill in the form below to add materials directly to the chatbot's knowledge base.

- Files will be processed and saved into the system's knowledge base.
- If you're pasting content, include a title, an optional URL, and the material content.
- For .txt files, only UTF-8 plain text is supported.

A. Please upload or submit only relevant, verified information in .txt or .html file

Upload a .txt or .html file

 Drag and drop file here
Limit 200MB per file • TXT, HTML, HTM

Browse files

B. Paste Materials Instead

Title (required)

URL (optional)

Material Content (required)

Submit Text

Existing Uploaded Files

 CommonTypesOfScams.txt

 HowtoReportScam.txt

File Content:

Tips to Avoid Being Scammed
No legitimate agency will ever ask for your SSN or to verify other personal information over the phone, unless it is for an application or issue that you have personally called them to initiate.
If you are shopping or banking online, make sure that the website is secure before giving out sensitive information. Look for the lock symbol in the browser. Don't give out sensitive information over unsecured networks.
Keep your passwords, PINs, and other important information private. If others have access to them, they can access your data and accounts.
No U.S. government agency will call you to ask for money. U.S. government agencies are likelier to contact you via regular mail if they have important information to send you/ask for.
No legitimate agency will ask you to purchase gift cards and provide them with the PIN.
Some scammers will threaten to take action against you unless you give them money. They may threaten to deport you, or they may threaten to take action against you on social media. This scare tactic is part of the scam. Never give someone money in this situation.
If it seems to good to be true, it probably is.
When in doubt, report it! It is okay to report something and be wrong

Download

Delete

 KeyScamIndicators.txt

 TipsToAvoidBeingScammed.txt

Figure 3: Data ingestion for internal ISSS use

Exploration of System Behavior

ScamBot is designed to serve as a virtual advisor, guiding users in evaluating suspicious communications. When a user inputs a message summary such as “I received a call saying my visa will be canceled unless I wire money today”, the bot will retrieve similar scam patterns from its knowledge base, flag it as likely fraudulent, and advises next steps (e.g., “do not share personal information,” “contact ISSS immediately”).

However, the generative nature of the model means it can occasionally behave unpredictably. The bot may overgeneralize and caution against legitimate communications or hallucinate unsupported details if the retrieved documents lack sufficient context. This may occur under the circumstance of explicitly instructing the model not to do so. These limitations indicate the need for further fine-tuning and expansion of the content base.

Limitations and Future Work

While ScamBot demonstrated effectiveness in identifying and responding to common scam scenarios, there are several limitations remain and present opportunities for future improvement. First, the chatbot currently cannot process image inputs, which significantly limits its ability to evaluate visual scam evidence such as fake checks, fake official letters, or screenshots of messages. In requiring users to manually describe the content of these images, this constraint reduces the model's usability and accuracy. However, if image ingestion features are included, it is important to consider stronger privacy safeguards as image files can contain data that may reveal personal information.

Second, the bot's current reliance on a ChatGPT 4.0 without finetuning means that its outputs can be overly cautious or imprecise, particularly when handling ambiguous scenarios. To improve its accuracy and judgment, future iterations should involve fine-tuning the model on a more robust set of real-world scam examples and incorporating user feedback.

Additionally, the scope of ScamBot could eventually be expanded beyond international students to serve a broader audience across campus, including faculty, staff, and domestic students. Building specialized tracks tailored to different scam types such as financial aid scams or phishing on social media would allow for more targeted guidance and enhance the bot's relevance across user groups.